

Passive expression pedal / I2C ADC

M1 | INDEPENDENT MODULE | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J302 / PASSIVE PEDAL
1: TIP I/O 2: RING I/O
3: GND 0V

JP301 / PEDAL JUMPER
1: TIP 2: RING
3: PEDAL IN 4: EXC OUT
5: RING 6: TIP

JP303 / DATA PULLUP
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C
1: 3V3 IN 2: GND 0V
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER
1: LOW GND 2: SELECT
3: HIGH 3V3

JP304 / CLOCK PULLUP
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT - LEAVE OPEN

J302
PASSIVE PEDAL INPUT / EXCITATION
OUTPUT

JP301
JUMPER ONLY: PEDAL POLARITY

R301

+3V3_A

1k / 1%

EXP_EXC

D303

BAT54H.115

+3V3_A

GND

D304

BAT54H.115

R302

10k / 1%

R303

1M / 1%

GND

C301

100n / 50V X7R

GND

R304

10k / 1%

EXP_EXC

D305

BAT54H.115

+3V3_A

C302

100n / 50V X7R

GND

D306

BAT54H.115

+3V3_A

U301

ADS1115IDGSR

ALERT/RDY

AIN0

AIN1

AIN2

AIN3

GND

GND

GND

GND

GND

GND

GND

GND

GND

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GND

R308

2.2k / 1%

+3V3_A

JP303

JUMPER ONLY: SDA PULLUP

SDA

R309

2.2k / 1%

+3V3_A

JP304

JUMPER ONLY: SCL PULLUP

SCL

R305

33R / 1%

SCL

R306

33R / 1%

SDA

J101

3V3 POWER INPUT + I2C TO HOST

+3V3

GND

GND

GND

JP302

JUMPER ONLY: ADC ADDRESS

GND

GND

GND

GND

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Engineering prototype – bench validation required

Ardor

Sheet: /

File: Ardor_Expression.kicad_sch

Title: Passive expression pedal / I2C ADC

Size: A3

Date: 2026-10-02

Rev: M1

KiCad E.D.A. 9.0.2

Id: 1/1

+3V3

+3V3_A

CHASSIS

GND